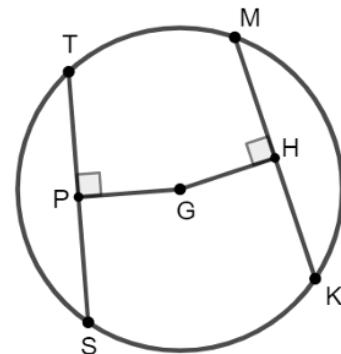


Geometry
Unit 12
Lesson 1 Practice

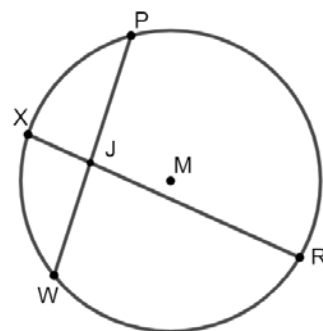
Name Answer Key
Date _____
Period _____

Circle G with points T, M, K , and S on the circle are shown with chords $\overline{TS}$ and $\overline{MK}$ equidistant from the center. $TS = (9x + 6)$, $MK = (11x - 10)$, and $m\angle TPG = m\angle MHG = 90^\circ$



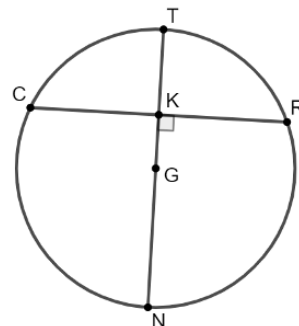
1. Find the value of x .
 $9x + 6 = 11x - 10$
 $16 = 2x$
 $x = 8$
2. What is the length of chord $\overline{MK}$?
 $MK = 11x - 10 = 11(8) - 10 = 78$

In Circle M , chords $\overline{RX}$ and $\overline{PW}$ intersect at point J . Points R, W, X , and P lie on Circle M . $RJ = (8x + 3)$, $JX = 4$, $PJ = (4x - 2)$, and $JW = 12$.



3. Solve for the value of x .
 $12(4x - 2) = 4(8x + 3)$
 $48x - 24 = 32x + 12$
 $16x = 36$
 $x = 2.25$
4. What is the length of line segment $\overline{RJ}$?
 $RJ = 8(2.25) + 3 = 21$
5. Find the length of chord $\overline{PW}$.
 $PW = 12 + 4(2.25) - 2 = 19$

Circle G is shown where points C, T, R , and N are on the circle. The diameter, $\overline{NT}$, is a perpendicular bisector of chord $\overline{CR}$. $CK = (5x + 6)$ and $KR = (7x - 2)$.



6. Calculate the value of x .
 $5x + 6 = 7x - 2$
 $-2x = -8$
 $x = 4$
7. What is the length of chord $\overline{CR}$?
 $CR = 5(4) + 6 + 7(4) - 2$
 $CR = 20 + 6 + 28 - 2$
 $CR = 52$

Chord $\overline{CM}$ is drawn in Circle B so that it intersects the diameter $\overline{JP}$.

8. Given that $CM = 24$ and $TB = 5$, what is the length of $\overline{BP}$?

$$5^2 + 12^2 = BM^2$$

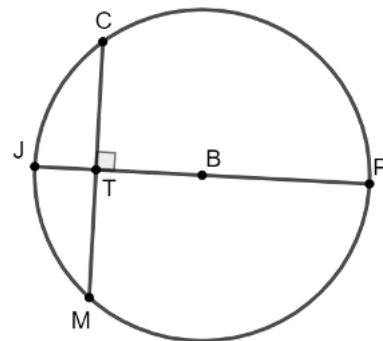
$$25 + 144 = BM^2$$

$$\sqrt{169} = \sqrt{(BM)^2}$$

$$13 = BM$$

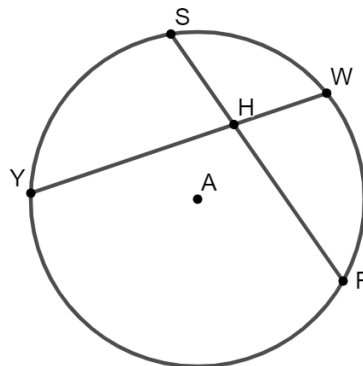
$\overline{BM}$ and $\overline{BP}$ are radii and therefore they are equal in length.

$$BM = BP = 13$$



Circle A is shown where points Y, S, W , and F lie on the circle. Chords $\overline{SF}$ and $\overline{YW}$ intersect at point H . $YH = (4x + 2)$, $SH = 9$, $HW = 8$, and $HF = (4x)$. Mia and Nazir are solving for x and their work is shown below.

Mia's Work	Nazir's Work
$8(4x + 2) = 9(4x)$ $32x + 2 = 36x$ $4x = 2$ $x = 0.5$	$8(4x + 2) = 9(4x)$ $32x + 16 = 36x$ $4x = 16$ $x = 4$



9. Whose work do you agree with? Justify your answer.

Nazir is correct. Mia did not distribute the 8 correctly to both terms in the parentheses.

In Circle H , points P, L, M , and S lie on the circle. Chord $\overline{PM}$ intersects the diameter $\overline{SL}$ at point G and $m\angle HGP = 90^\circ$.

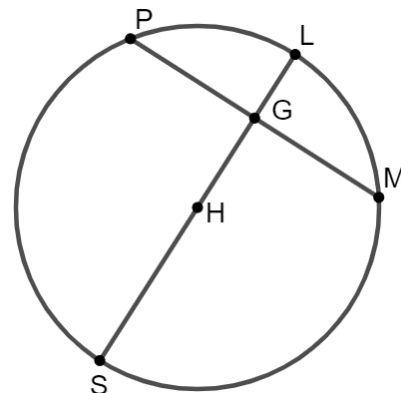
10. If $PM = 32$ and $HS = 18$, what is the length of $\overline{GH}$? Round to the nearest hundredth if necessary.

$$18^2 = 16^2 + GH^2$$

$$324 = 256 + GH^2$$

$$\sqrt{68} = \sqrt{(GH)^2}$$

$$8.25 \approx GH$$



Geometry
Unit 12
Lesson 2 Practice

Name Answer Key
Date _____
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Circle A with secants $\overline{PM}$ and $\overline{CM}$ is shown. $\overline{PM}$ and $\overline{CM}$ intersect at point M outside the circle so that $PD = 17$, $DM = 3$, $CK = x$, and $KM = 5$.

1. What is the length of secant $\overline{PM}$?

$$PM = PD + DM = 17 + 3 = 20$$

2. Find the value of x .

$$5(x + 5) = 3(20)$$

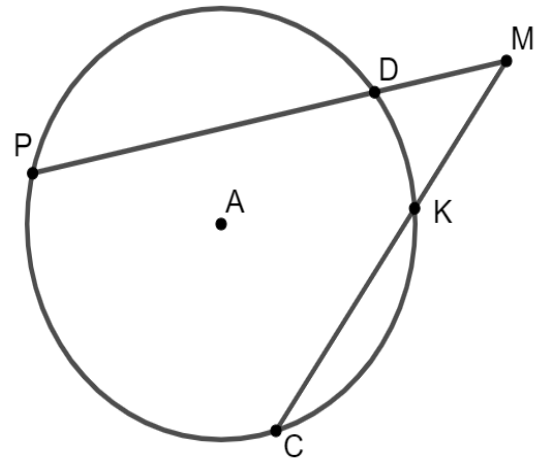
$$5x + 25 = 60$$

$$5x = 35$$

$$x = 7$$

3. What is the length of secant $\overline{CM}$?

$$CM = CK + KM = 7 + 5 = 12$$



Circle L is shown, where points G , X , A , and H lie on circle L . Secants $\overline{BX}$ and $\overline{BA}$ intersect at point B outside the circle such that $BG = 8$, $GX = (7x - 4)$, $BH = 6$, and $HA = 18$.

4. What is the value of x ?

$$8(8 + 7x - 4) = 6(6 + 18)$$

$$8(7x + 4) = 144$$

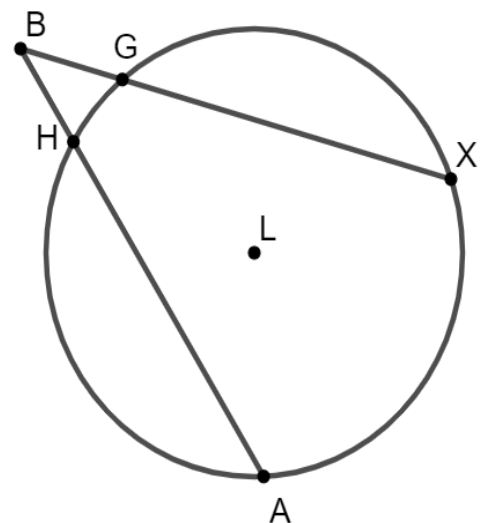
$$56x + 32 = 144$$

$$56x = 112$$

$$x = 2$$

5. Calculate the length of secant $\overline{BX}$.

$$BX = BG + GX = 8 + 7(2) - 4 = 18$$



Circle M with secants $\overline{GT}$ and $\overline{ST}$ is shown. Secants $\overline{GT}$ and $\overline{ST}$ intersect outside the circle at point T so that $GW = 7x$, $WT = 2x$, $SN = 6$, and $NT = 6$.

6. Find the value of x .

$$2x(7x + 2x) = 6(6 + 6)$$

$$14x^2 + 4x^2 = 72$$

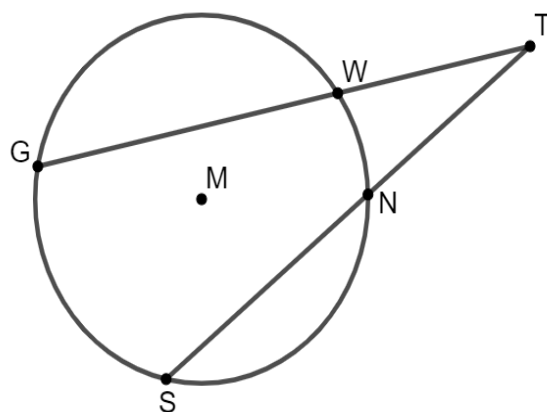
$$18x^2 = 72$$

$$x^2 = 4$$

$$x = 2$$

7. What is the length of secant $\overline{GT}$?

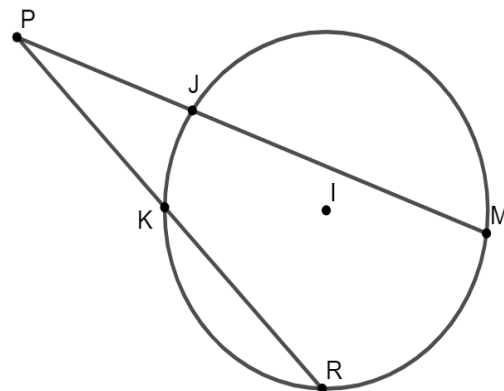
$$GT = 7x + 2x = 9(2) = 18$$



Bailey and Luciano were solving for x in circle I where points J , M , R , and K are on the circle and secants $\overline{PM}$ and $\overline{PR}$ intersect at point P outside the circle. They know $PJ = 2$, $JM = (12x - 5)$, $PK = 3$, and $KR = 4x$.

Each of their equations are set up to solve for x .

Bailey's Work	Luciano's Work
$3(4x + 3) = 2(12x - 3)$	$4x(4x + 3) = 2(12x - 3)$



8. Who wrote the correct equation needed to solve for x ?

Bailey

9. Use this equation to find the value of x .

$$3(4x + 3) = 2(12x - 3)$$

$$12x + 9 = 24x - 6$$

$$-12x = -15$$

$$x = \frac{15}{12} = \frac{5}{4} = 1.25$$

10. What is the length of secant $\overline{PR}$?

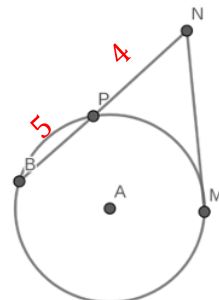
$$PR = PK + KR = 3 + (4x) = 3 + 4(1.25) = 3 + 5 = 8$$

Geometry
Unit 12
Lesson 3 Practice

Name _____
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Period _____

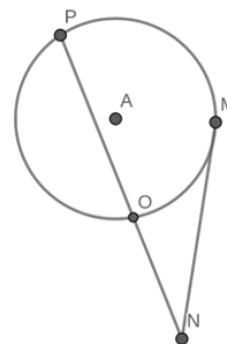
1. Circle A is shown with points M, B and P on the circle. Secant $\overline{BN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PB = 5$ and $PN = 4$. What is the length of $\overline{MN}$?

$$\begin{aligned}PN(BP + PN) &= MN^2 \\4(5 + 4) &= MN^2 \\4(9) &= MN^2 \\\sqrt{36} &= \sqrt{MN^2} \\MN &= 6\end{aligned}$$



2. Circle A is shown with points M, O and P on the circle. Secant $\overline{PN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PO = (3x + 3)$, $ON = 10$ and $MN = 20$. Find the value of x .

$$\begin{aligned}MN^2 &= 10(10 + 3x + 3) && \rightarrow 270 = 30x \\20^2 &= 130 + 30x && 9 = x \\400 &= 130 + 30x && x = 9\end{aligned}$$

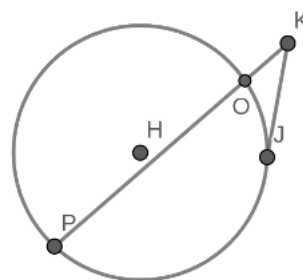


3. What is the length of $\overline{PN}$?

$$\begin{aligned}PN &= 13 + 3(9) \\&= 40 \\PN &= 40\end{aligned}$$

4. Circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 112$, $OK = 14$ and $KJ = x$. Find the value of x .

$$\begin{aligned}OK(PO + OK) &= KJ^2 \\14(112 + 14) &= x^2 \\14(126) &= x^2 \\\sqrt{1764} &= \sqrt{x^2} \\x &= 42\end{aligned}$$

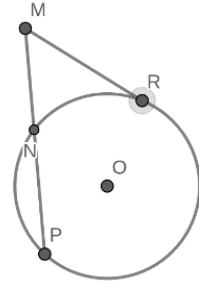


5. Using the same image as question 4, circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 16 + x$, $OK = 2$ and $KJ = 8$. Find the length of $\overline{KP}$.

$$\begin{aligned}2(16 + x + 2) &= 8^2 && KP = PO + OK \\2(18 + x) &= 64 && 16 + 14 + 2 \\36 + 2x &= 64 && KP = 32 \\2x &= 28 \\x &= 14\end{aligned}$$

6. Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = 20$, $MN = 16$ and $MR = 4x$. What is the value of x ?

$$\begin{aligned} MN(PN + MN) &= MR^2 \\ 16(20 + 16) &= (4x)^2 \\ 576 &= 16x^2 \\ 36 &= x^2 \\ x &= 6 \end{aligned}$$

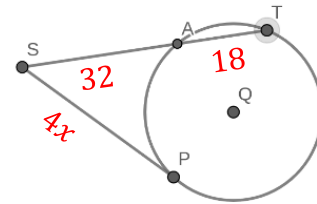


7. Using the same image as question 6, Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = x + 10$, $MN = 15$ and $MR = 30$. What is the value of x ?

$$\begin{aligned} MN(PN + MN) &= MR^2 \\ 15(x + 10 + 15) &= 30^2 \\ 15x + 375 &= 900 \end{aligned} \quad \begin{aligned} &\rightarrow 15x = 525 \\ &x = 35 \end{aligned}$$

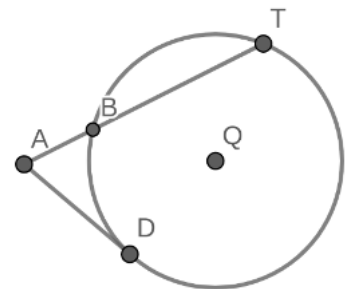
8. Circle Q is shown with points A, T and P on the circle. Secant $\overline{ST}$ and tangent $\overline{SP}$ intersect at point S outside the circle. $SP = 4x$, $AT = 18$ and $AS = 32$. What is the value of x ?

$$\begin{aligned} PS^2 &= AS(AS + TA) \\ (4x)^2 &= 32(18 + 32) \\ 16x^2 &= 1600 \\ x^2 &= 100 \\ x &= 10 \end{aligned}$$



9. Daniella and Taylor are working on finding the length of $\overline{AD}$. Circle Q is shown with points B, T and D on the circle. Secant $\overline{AT}$ and tangent $\overline{AD}$ intersect at point A outside the circle. $BT = 12$, $AB = 10$ and $AD = 2x$. Their partial work solving for x is shown below. Find the value of x for each of their work. Round to the nearest hundredth. Whose work do you agree with? **Daniella**

Daniella's work	Taylor's work
$(2x)^2 = 10(10 + 12)$ $4x^2 = 10(22)$ $x^2 = 55$ $x = 7.42$	$(2x)^2 = 12(10 + 12)$ $4x^2 = 12(22)$ $x^2 = 132$



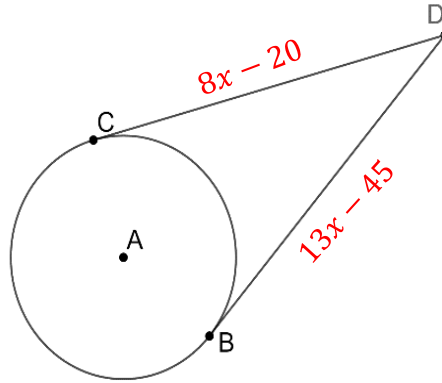
10. Using the image in Question 9, what is the length of $\overline{AD}$?

$$\begin{aligned} AD &= 2x \\ &= 2 \cdot 7.42 \\ AD &= 14.84 \end{aligned}$$

Geometry
Unit 12
Lesson 4 Practice

Name **Answer Key** _____
Date _____
Period _____

For questions 1-3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 8x - 20$ and $DB = 13x - 45$.



1. What is the value of x ?

$$\begin{aligned} 8x - 20 &= 13x - 45 \\ -5x &= -25 \\ \boxed{x = 5} \end{aligned}$$

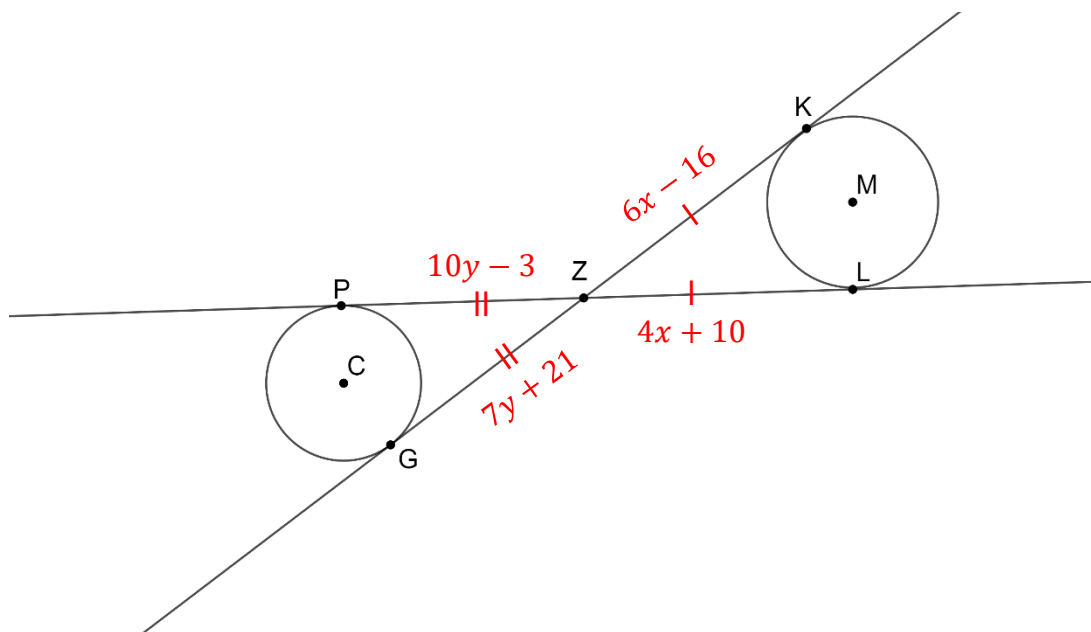
2. What is the length of $\overline{CD}$?

$$\begin{aligned} CD &= 8(5) - 20 \\ \boxed{CD = 20} \end{aligned}$$

3. What is the length of $\overline{DB}$?

$$\begin{aligned} DB &= 13(5) - 45 \\ 65 - 45 \\ \boxed{DB = 20} \end{aligned}$$

For questions 4-10, Circles C and M are shown. Lines $\overline{PL}$ and $\overline{GK}$ intersect at point Z . Line $\overline{PL}$ is tangent to circle C at point P and circle M at point L . Line $\overline{GK}$ is tangent to circle C at point G and circle M at point K . $PZ = 10y - 3$, $ZL = 4x + 10$, $GZ = 7y + 21$, $ZK = 6x - 16$.



4. Write an equation you can use to solve for x .

$$4x + 10 = 6x - 16$$

5. Solve the equation you wrote in question 4 for x .

$$10 = 2x - 16$$

$$26 = 2x$$

$$x = 13$$

6. Write an equation you can use to solve for y .

$$10y - 3 = 7y + 21$$

7. Solve the equation you wrote in question 6 for y .

$$3y - 3 = 21$$

$$3y = 24$$

$$y = 8$$

8. What is the length of $\overline{GK}$?

$$GK = GZ + ZK \quad (7 \cdot 8 + 21) + (6 \cdot 13 - 16)$$

$$77 + 62$$

$$GK = 139$$

9. What is the length of $\overline{GZ}$?

$$GZ = 7(8) + 21$$

$$GZ = 77$$

10. What is the length of $\overline{ZL}$?

$$ZL = 4(13) + 10$$

$$= 52 + 10$$

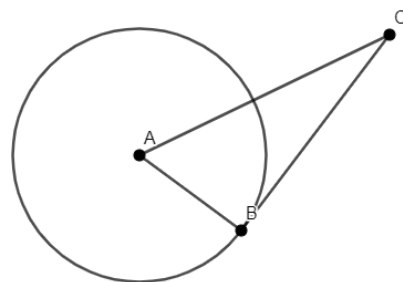
$$ZL = 62$$

Geometry
Unit 12
Lesson 5 Practice

Name **Answer Key** _____
Date _____
Period _____

1. Circle A is shown with point C outside the circle, where point B is on the circle, $AC = 26$, $CB = 24$, and $BA = 10$. Verify if $\overline{BC}$ is tangent to circle A using Pythagorean Theorem.

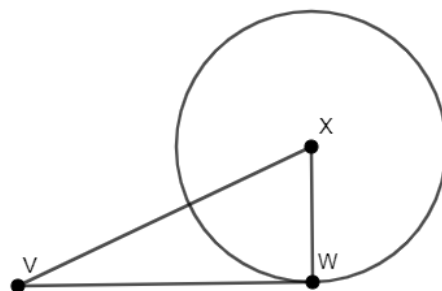
$$\begin{aligned}(24)^2 + (10)^2 &= (26)^2 \\ 576 + 100 &= 676 \\ 676 &= 676 \checkmark \Rightarrow \text{TANGENT}\end{aligned}$$



Circle X on the right has $\overline{VW}$ as a tangent segment where W is the point of tangency. $VX = 17$ and $VW = 15$.

2. What is the length of $\overline{XW}$?

$$\begin{aligned}(15)^2 + (XW)^2 &= (17)^2 \\ 225 + (XW)^2 &= 289 \\ (XW)^2 &= 64 \Rightarrow XW = 8\end{aligned}$$



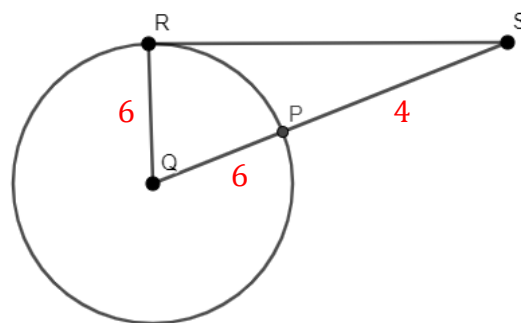
3. What is the area of $\triangle XWV$?

$$\frac{1}{2}(15)(8) = 60 \text{ units}^2$$

In Circle Q , $\overline{RS}$ is tangent to the circle at point R , $QR = 6$, and $PS = 4$.

4. What is the length of $\overline{RS}$?

$$\begin{aligned}6^2 + (RS)^2 &= (10)^2 \\ 36 + (RS)^2 &= 100 \\ (RS)^2 &= 64 \Rightarrow RS = 8\end{aligned}$$



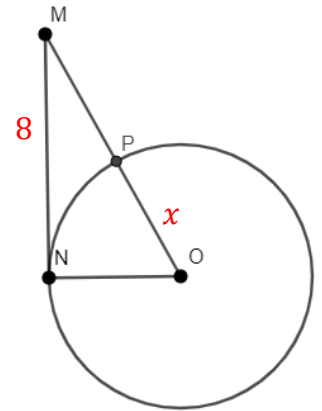
5. What is the area of $\triangle QRS$?

$$\frac{1}{2}(6)(8) = 24 \text{ units}^2$$

In Circle O , $\overline{MN}$ is a tangent segment whose point of tangency is point N . $MN = 8$ and $MP = 2$.

6. What is the length of $\overline{ON}$?

$$\begin{aligned}(2 + x)^2 - 8^2 &= x^2 \\ 4 + 4x + x^2 - 64 &= x^2 \\ 4x &= 60 \\ x &= 15 \\ ON &= 15\end{aligned}$$



7. What is the length of $\overline{MO}$?

$$\begin{aligned}2 + 15 &= 17 \\ MO &= 17\end{aligned}$$

8. What is the perimeter of $\triangle MNO$?

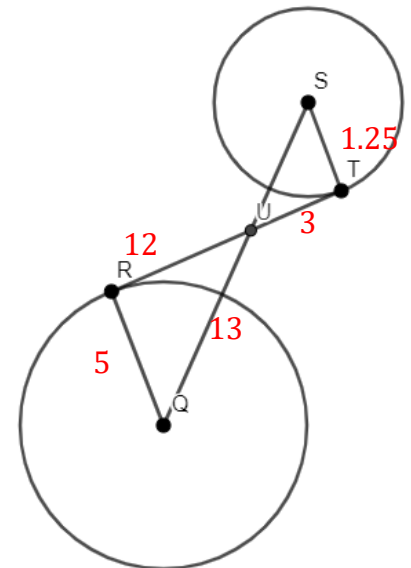
$$17 + 15 + 8 = 40 \text{ units}$$

In the figure on the right, $\overline{RT}$ is a common tangent segment of both Circle Q and Circle S . $QR = 5$, $RU = 12$, and $TU = 3$.

9. What is the length of $\overline{ST}$?

$$\frac{5}{12} = \frac{ST}{3}$$

$$15 = 12(ST) \Rightarrow \boxed{ST = 1.25}$$



10. What is the length of $\overline{QS}$?

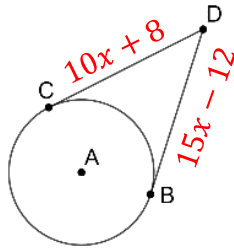
$$\begin{aligned}(12)^2 + (5)^2 &= (QU)^2 \\ 13 &= QU\end{aligned}$$

$$\begin{aligned}(SU)^2 &= 3^2 + (1.25)^2 & \Rightarrow QS &= 13 + 3.25 \\ SU &= 3.25 & QS &= 16.25\end{aligned}$$

Geometry
Unit 12
Lesson 6 Practice

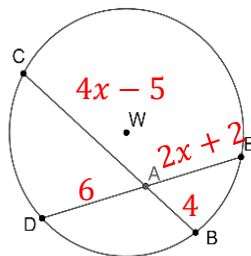
Name **Answer Key** _____
Date _____
Period _____

For questions 1 through 3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 10x + 8$ and $DB = 15x - 12$.



1. What is the value of x ?
 $10x + 8 = 15x - 12$
 $20 = 5x$
 $4 = x$
2. What is the length of $\overline{CD}$?
 $10(4) + 8 = 40 + 8 = 48$
3. What is the length of $\overline{DB}$?
 $15(4) - 12 = 60 - 12 = 48$

For questions 4 through 7, Circle W is shown. Chords $\overline{CB}$ and $\overline{DE}$ intersect at point A .
 $CA = 4x - 5$, $AB = 4$, $DA = 6$, and $AE = 2x + 2$.



4. Write an equation you can use to solve for x .
 $4(4x - 5) = 6(2x + 2)$
 $16x - 20 = 12x + 12$

5. Solve the equation you wrote in question 4 for x .

$$16x - 20 = 12x + 12$$

$$4x = 32$$

$$x = 8$$

6. What is the length of $\overline{CB}$?

$$4(8) - 5 = 32 - 5 = 27$$

$$\overline{CB} = 27 + 4 = 31$$

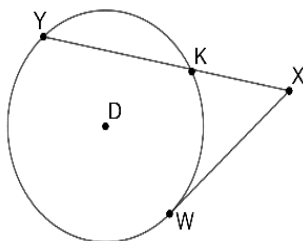
7. What is the length of $\overline{DE}$?

$$2(8) + 2 = 16 + 2 = 18$$

$$\overline{DE} = 6 + 18 = 24$$

For questions 8 through 10, Circle D is shown. $\overline{XY}$ is a secant to the circle and intersects tangent $\overline{XW}$ at point X . $XW = 10$, $KX = 4$, and $YX = 5x + 10$.

$$KX(KX + YK) = (WX)^2$$



8. Write an equation that can be used to find the value of x .

$$4(5x + 10) = (10)^2$$

9. Solve the equation you wrote in question 8 to find the value of x .

$$20x + 40 = 100$$

$$20x = 60$$

$$x = 3$$

10. What is the length of $\overline{YK}$?

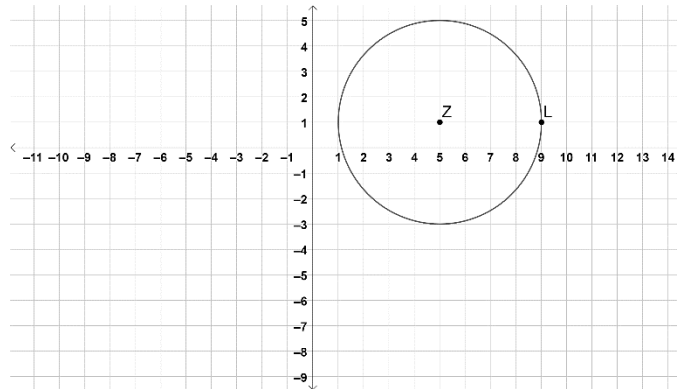
$$YX = 5(3) + 10 = 15 + 10 = 25$$

$$25 - 4 = 21 \Rightarrow YK = 21$$

Geometry
Unit 12
Lesson 7 Practice

Name **Answer Key** _____
Date _____
Period _____

For questions 1 and 2, Circle Z is shown. Point L lies on the circle.



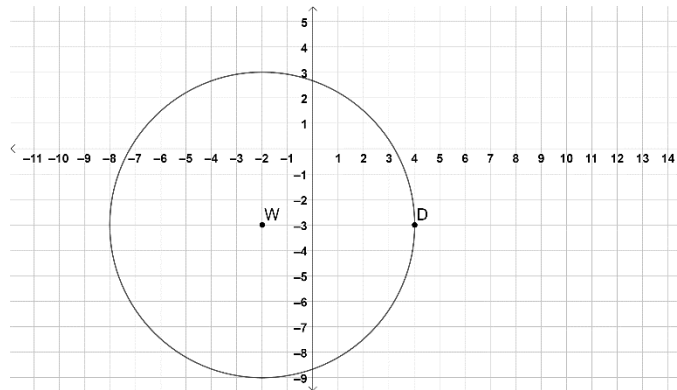
1. Give the key features of the circle.

Center	Diameter
(5, 1)	8

2. Write the equation that represents Circle Z .

$$(x - 5)^2 + (y - 1)^2 = 16$$

For questions 3 and 4, Circle W is shown. Point D lies on the circle.



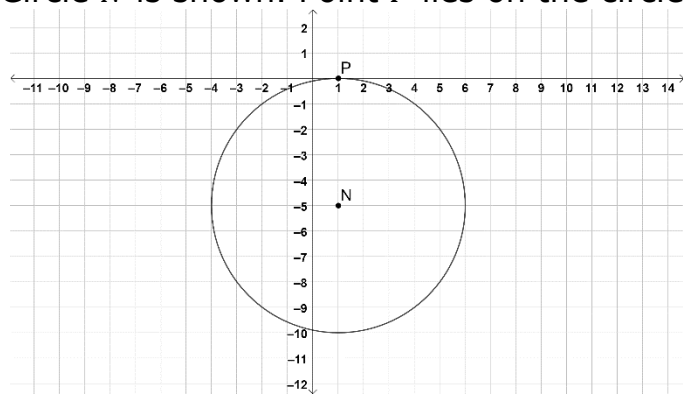
3. Give the key features of the circle.

Center	Diameter
(-2, -3)	12

4. Write the equation that represents Circle W .

$$(x + 2)^2 + (y + 3)^2 = 36$$

For questions 5 and 6, Circle *N* is shown. Point *P* lies on the circle.



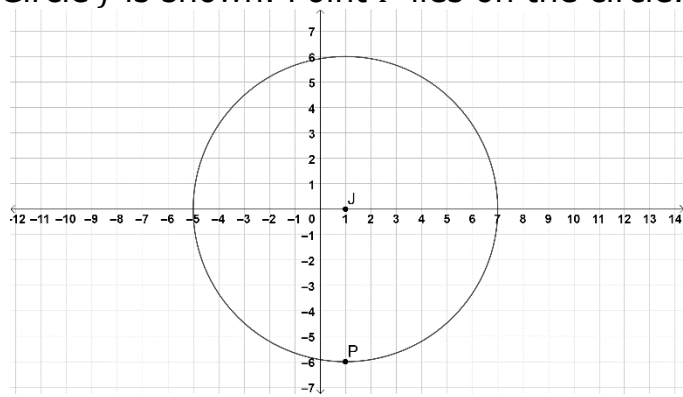
5. Give the key features of the circle.

Center	Diameter
(1, -5)	10

6. Write the equation that represents Circle *N*.

$(x - 1)^2 + (y + 5)^2 = 25$

For questions 7 and 8, Circle *J* is shown. Point *P* lies on the circle.



7. Give the key features of the circle.

Center	Diameter
(1, 0)	12

8. Write the equation that represents Circle *J*.

$(x - 1)^2 + y^2 = 36$

Geometry
Unit 12
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

For questions 1 and 2, consider the equation of the circle represented by
 $(x - 8)^2 + (y - 10)^2 = 12^2$.

1. Rewrite the equation in general form.

$$\begin{aligned}(x - 8)^2 + (y - 10)^2 &= 12^2 \\ x^2 - 16x + 64 + y^2 - 20y + 100 &= 144 \\ x^2 + y^2 - 16x - 20y + 20 &= 0\end{aligned}$$

2. Give the key features of the circle.

Center	Diameter	Radius
(8, 10)	24	12
Domain	Range	
$-4 \leq x \leq 20$	$-2 \leq y \leq 22$	

For questions 3 and 4, consider the equation of the circle represented by
 $(x + 10)^2 + (y - 3)^2 = 1^2$.

3. Rewrite the equation in general form.

$$\begin{aligned}(x + 10)^2 + (y - 3)^2 &= 1^2 \\ x^2 + 20x + 100 + y^2 - 6y + 9 &= 1 \\ x^2 + y^2 + 20x - 6y + 108 &= 0\end{aligned}$$

4. Give the key features of the circle.

Center	Diameter	Radius
(-10, 3)	2	1
Domain	Range	
$-11 \leq x \leq -9$	$2 \leq y \leq 4$	

For questions 5 and 6, consider the equation of the circle represented by
 $x^2 + y^2 - 18y = -65$.

5. Rewrite the equation in center-radius form.

$$\begin{aligned}x^2 + y^2 - 18y &= -65 \\ x^2 + y^2 - 18y + 81 &= -65 + 81 \\ x^2 + (y - 9)^2 &= 16\end{aligned}$$

6. Give the key features of the circle.

Center	Diameter	Radius
(0, 9)	8	4
Domain	Range	
$-4 \leq x \leq 4$	$5 \leq y \leq 13$	

For questions 7 and 8, consider the equation of the circle represented by

$$x^2 + y^2 + 10x = -21.$$

7. Rewrite the equation in center-radius form.

$$\begin{aligned} x^2 + 10x + y^2 &= -21 \\ x^2 + 10x + 25 + y^2 &= -21 + 25 \\ (x + 5)^2 + y^2 &= 4 \end{aligned}$$

8. Give the key features of the circle.

Center	Diameter	Radius
$(-5, 0)$	4	2
Domain	Range	
$-7 \leq x \leq -3$	$-2 \leq y \leq 2$	

For questions 9 and 10, consider the equation of the circle represented by

$$x^2 + y^2 - 2x + 10y = 23.$$

9. Rewrite the equation in center-radius form.

$$\begin{aligned} x^2 - 2x + y^2 + 10y &= 23 \\ x^2 - 2x + 1 + y^2 + 10y + 25 &= 23 + 1 + 25 \\ (x - 1)^2 + (y + 5)^2 &= 49 \end{aligned}$$

10. Give the key features of the circle.

Center	Diameter	Radius
$(1, -5)$	14	7
Domain	Range	
$-6 \leq x \leq 8$	$-12 \leq y \leq 2$	

Geometry
Unit 12
Lesson 9 Practice

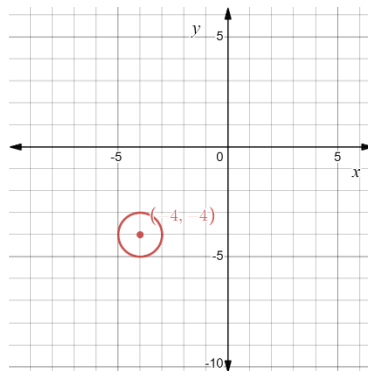
Name Answer Key
Date _____
Period _____

For questions 1 and 2, consider that the equation of circle B is $(x + 4)^2 + (y + 4)^2 = 1$.

1. Give the key features of the circle.

Center	Radius
$(-4, -4)$	1
Domain	Range
$-5 \leq x \leq -3$	$-5 \leq y \leq -3$

2. Graph circle B .

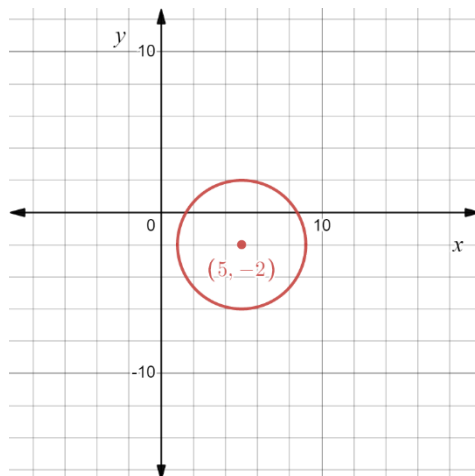


For questions 3 and 4, consider that the equation of circle J is $(x - 5)^2 + (y + 2)^2 = 16$.

3. Give the key features of the circle.

Center	Radius
$(5, -2)$	4
Domain	Range
$1 \leq x \leq 9$	$-6 \leq y \leq 2$

4. Graph circle J .



For questions 5 through 7, consider that the equation of circle B is

$$x^2 - 10x + y^2 - 6y = 30.$$

5. Rewrite the equation of the circle in center-radius form.

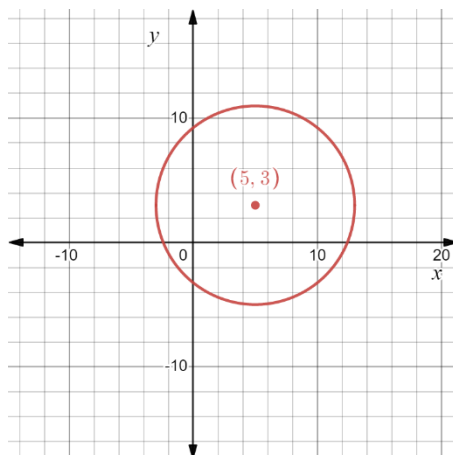
$$x^2 - 10x + 25 + y^2 - 6y + 9 = 30 + 25 + 9$$

$$(x - 5)^2 + (y - 3)^2 = 64$$

6. Give the key features of the circle.

Center	Radius
$(5, 3)$	8
Domain	Range
$-3 \leq x \leq 13$	$-5 \leq y \leq 11$

7. Graph circle B .



For questions 8 through 10, consider that the equation of circle Y is

$$x^2 + 2x + y^2 - 10y + 17 = 0.$$

8. Rewrite the equation of the circle in center-radius form.

$$x^2 + 2x + 1 + y^2 - 10y + 25 = -17 + 1 + 25$$

$$(x + 1)^2 + (y - 5)^2 = 9$$

9. Give the key features of the circle.

Center	Radius
$(-1, 5)$	3
Domain	Range
$-4 \leq x \leq 2$	$2 \leq y \leq 8$

10. Graph circle Y .

